## 0 · L00: AI & ML basics

Four pillars: linear algebra, geometry, calculus, probability. AI coined **1956 (McCarthy)**; **Turing 1950** imitation game (30% fool for 5 min by 2000); Deep Blue 1997; deep learning 2012. Traditional: data + program → output. **ML: data + output → program.** Data → **algorithm** (we design) → **model** $f(x)$ (LR: coefficient vector; tree: if-then; NN: trained weights).

**Supervised** (labelled): regression = continuous ("**probability** prices fall"), classification = discrete ("**whether** prices fall"). **Unsupervised** (unlabelled, clustering, market segmentation). **Semi-supervised**: fit on labelled, label the rest, refit.

Features (inputs), label (output), training data, loss/cost, test data. MNIST 28×28, 60k train/10k test. Flow: problem nature → collect data → train → validate → deploy.

CITY lesson: "math-informed AI — decompose first, then learn on the pieces."

## 1 · Vectors: notation & combinations

$x \in \mathbb{R}^n$: ordered list, size $n$, entry $x_i$ (index from 1). Equal ⇔ same size & all entries equal.

Special: $\mathbf{0}$, $\mathbf{1}$, $e_i$ ($(e_i)_j = \delta_{ij}$); $x = \sum_i x_i e_i$. **$k$-sparse**: at most $k$ non-zeros; $\|x\|_0 = \#\text{non-zeros}$ (**not** a norm).

Readings: list / point / displacement ($\Delta x = x' - x$).

Encodings: Boolean $\pm 1$: $y = 2x - \mathbf{1}$, $x = \frac{1}{2}(y + \mathbf{1})$; word counts: $\mathbf{1}^\top w$=total, $h = w/\mathbf{1}^\top w$; currency value $\rho^\top c$.

$u + v$, $\alpha v$ entry-wise; same size needed. $x^2$ is meaningless for $n > 1$; $w^\top z$ (col×row) = outer product.

$y = \sum c_i v_i$: **linear** (any $c$) ⊃ **affine** ($\sum c_i = 1$) ⊃ **convex** ($\sum c_i = 1$, $c_i \geq 0$); **conic**: $c_i \geq 0$.

Two points: linear → span; conic → wedge; affine → **line** through them ($x_2 + t(x_1 - x_2)$); convex → **segment**. 3 non-collinear points, affine → plane.

$2x_A - x_B$: linear & affine, not convex. Combination of combinations is a combination.

## 2 · Inner product

$u^\top v = \sum u_i v_i = \|u\|\|v\| \cos\theta$; symmetric, linear; $u^\top u = \|u\|^2 \geq 0$. $u^\top v > 0$ acute, $= 0$ ⊥, $< 0$ obtuse.

Queries: $e_i^\top x = x_i$; $\mathbf{1}^\top x$=sum; $\frac{1}{n}\mathbf{1}^\top x$=mean; indicator $g^\top x$ =sum of selected entries; avg age $a^\top x / \mathbf{1}^\top x$.

Boolean $x, y$: $x^\top y$=common; $\|x\|_2^2 = \mathbf{1}^\top x$ (so $\|x\|_2$ is a square root!); $\mathbf{1}^\top(x - y)$=difference of counts.

Uses: score $\hat{y} = w^\top x + b$, neuron $z = w^\top x + b$, rating $u^\top v$, value $c^\top q$, similarity.

## 3 · Norms, distance, error

Norm axioms: $\|v\| \geq 0$; $\|v\| = 0 \Leftrightarrow v = \mathbf{0}$; $\|\alpha v\| = |\alpha|\|v\|$; $\|u + v\| \leq \|u\| + \|v\|$.

$\|v\|_p = (\sum |v_i|^p)^{1/p}$, $p \geq 1$. $\|v\|_1 = \sum |v_i|$, $\|v\|_2 = \sqrt{v^\top v}$, $\|v\|_\infty = \max |v_i|$; $\|v\|_\infty \leq \|v\|_2 \leq \|v\|_1$.

Unit balls in $\mathbb{R}^2$: $\ell_1$ diamond $(\pm 1, 0), (0, \pm 1)$; $\ell_2$ circle; $\ell_\infty$ square $(\pm 1, \pm 1)$.

$\|u \pm v\|^2 = \|u\|^2 \pm 2u^\top v + \|v\|^2$. ⊥ ⇒ Pythagoras; $\|\alpha a + \beta b\| = \sqrt{\alpha^2\|a\|^2 + \beta^2\|b\|^2}$. Unit $u, v$: $\|u - v\|^2 = 2(1 - u^\top v)$.

**Cauchy–Schwarz** $|u^\top v| \leq \|u\|\|v\|$ (= iff parallel). $\|(x, x)\| = \sqrt{2}\|x\|$; $\|x\| = -\|-x\|$ only if $x = \mathbf{0}$.

$\hat{u} = u/\|u\|$; opposite unit vector $-u/\|u\|$.

Distance $d_p = \|u - v\|_p$. RMSE $= \|y - \hat{y}\|_2/\sqrt{n}$; MAE $= \|y - \hat{y}\|_1/n$; $\ell_\infty$ = worst error.

## 4 · Angle, projection, statistics

- $\cos\theta = \dfrac{a^\top b}{\|a\|\|b\|}$ (cosine similarity; ignores size). $y = 3x \Rightarrow$ cos-sim 1 but $\|y - x\| = 2\|x\|$.
- $\text{proj}_u\, x = \dfrac{x^\top u}{u^\top u} u$ $\left(= (x^\top u)u \text{ if } \|u\| = 1\right)$; residual $x - \text{proj} \perp u$.
- $\bar{x} = \frac{\mathbf{1}^\top x}{n}$, $\text{rms} = \frac{\|x\|}{\sqrt{n}}$, $\text{std} = \frac{\|x - \bar{x}\mathbf{1}\|}{\sqrt{n}}$ (divide by $n$); $\text{rms}^2 = \text{avg}^2 + \text{std}^2$.
- Centred $x_c = x - \bar{x}\mathbf{1} \perp \mathbf{1}$. **Correlation** $\rho = \dfrac{\tilde{a}^\top \tilde{b}}{\|\tilde{a}\|\|\tilde{b}\|} \in [-1, 1]$ = cosine of centred vectors; $y = \alpha x + \beta\mathbf{1} \Rightarrow \rho = \text{sign}\,\alpha$.
- Standardise $z = (x - \bar{x}\mathbf{1})/\,\text{std}$ (mean 0, std 1) before kNN, k-means, SVM, GD.
- **Bias–variance:** RMS err$^2$ = Bias$^2$+Var+Noise. Bias = too simple → underfit; variance = too sensitive → overfit.

## 5 · Span, independence, basis

- $\text{span}(v_i) = \{\sum c_i v_i\}$ — always a subspace. $b \in$ span ⇔ $[v_1 \cdots v_k]c = b$ solvable. 2 non-parallel vectors in $\mathbb{R}^3$ → plane; parallel → line.
- **Independent:** $\sum c_i v_i = \mathbf{0} \Rightarrow$ all $c_i = 0$. Dependent ⇔ one is a combination of others. Set with $\mathbf{0}$ is dependent; "no zero vector" ⇏ independent; square: dependent ⇔ $\det = 0$.
- **Independence–dimension:** in dim $n$, at most $n$ independent; $n + 1$ always dependent; spanning set needs $\geq n$; $n$ independent in $\mathbb{R}^n \Rightarrow$ basis.
- **Basis** = independent + spanning ⇒ unique coordinates. Need not be orthogonal ($\{(1, 0), (1, 1)\}$).
- **Orthonormal**: $q_i^\top q_j = \delta_{ij}$; ⊥ non-zero ⇒ independent; coefficients $c_i = q_i^\top u$. Membership test: $\hat{a} = \sum (q_i^\top a) q_i = a$?

## 6 · Gram–Schmidt

- $\tilde{q}_i = a_i - \sum_{j<i} (q_j^\top a_i) q_j$, $q_i = \tilde{q}_i / \|\tilde{q}_i\|$. $\tilde{q}_i = \mathbf{0} \Rightarrow a_i$ dependent (dim of span = #non-zero $\tilde{q}$).
- Keeps span at every step; output **depends on order**; works in any $\mathbb{R}^n$; take a unit-length input first. Basis of QR.
- Ex: $(4, 3), (-2, 1) \to q_1 = (\frac{4}{5}, \frac{3}{5})$, $q_1^\top a_2 = -1$, $\tilde{q}_2 = (-\frac{6}{5}, \frac{8}{5})$, $q_2 = (-\frac{3}{5}, \frac{4}{5})$. Always **check** $q_1^\top q_2 = 0$.
- Span of 2 vectors in $\mathbb{R}^3$: plane with normal $u_1 \times u_2$.

## 7 · Vector spaces & subspaces

- 10 axioms (closure $+$ and $\cdot$, comm., assoc., $\mathbf{0}$, $-u$, distributive ×2, $(ab)u = a(bu)$, $1u = u$). Examples: $\mathbb{R}^n$, polynomials, functions. $\mathbb{R}^4_{\geq 0}$ is **not**.
- **Subspace** ⇔ $\mathbf{0} \in W$, closed under $+$ and scalar $\cdot$ ($\alpha u + \beta v \in W$). Lines/planes **through 0**: yes. Line not through 0, quadrant, line∪plane: no.
- dim = #vectors in any basis; $\dim\{\mathbf{0}\} = 0$. Subspace properties ≠ "dim ≥ 2".

## 8 · Matrices: basics

- $A \in \mathbb{R}^{m \times n}$, $A_{ij}$ row $i$ col $j$. Square/tall ($m > n$)/wide; $0$, $I$ ($\delta_{ij}$), diagonal, triangular, block, symmetric, orthogonal, permutation.
- $A^\top$: $(A^\top)_{ij} = A_{ji}$; $(AB)^\top = B^\top A^\top$, $(ABC)^\top = C^\top B^\top A^\top$.
- **Symmetric** $A = A^\top$; **skew** $A = -A^\top$ (diag $= 0$). $A = S + K$, $S = \frac{1}{2}(A + A^\top)$, $K = \frac{1}{2}(A - A^\top)$. $A^2$ sym if $A$ sym; $AB$ sym iff $AB = BA$; $A^\top A$, $AA^\top$ always sym.
- $\|A\|_F = \sqrt{\sum A_{ij}^2}$; $\|A\|_F^2 = \text{tr}(A^\top A)$. $\text{tr}\, A = \sum A_{ii}$, linear, $\text{tr}\, A^\top = \text{tr}\, A$, $\text{tr}\, I_n = n$, $\text{tr}(AB) = \text{tr}(BA)$, cyclic; $\text{tr}(AB) \neq$

$\operatorname{tr} A \operatorname{tr} B$; $\operatorname{tr} A = \sum \lambda_i$.

## 9 · Products

$(Ax)_i = \sum_k A_{ik} x_k$ (cols of $A$ = size of $x$). Views: **rows** (dot products), **columns** $Ax = \sum_j x_j a_j$, **linear map** $A(x + x') = Ax + Ax'$.

$Ae_j = a_j$; all $Ax$ = column space; dependent columns $\Leftrightarrow$ $Ac = \mathbf{0}$, $c \neq \mathbf{0}$. Permutation $P$: one 1 per row/col, $P^{-1} = P^\top$.

Applications: $Ap$ stock value, $Aw$ weight; $A^\top w$ scores; $A^\top q$ similarity; $z = Wx$ classifier (argmax); $\hat{y} = Xw + b\mathbf{1}$.

$(AB)_{ij}$ = row $i$ of $A$ · col $j$ of $B$; $(m \times n)(n \times p) = m \times p$.

Not commutative; associative; distributive; $AB = [Ab_1 \cdots Ab_p]$; blocks multiply like entries.

**Outer product** $ab^\top$ ($m \times n$, entries $a_i b_j$): rank 1. **Gram** $G = A^\top A$, $G_{ij} = a_i^\top a_j$, symmetric.

Linear layers collapse: $W^{[2]} W^{[1]} x = Wx$ — no non-linearity $\Rightarrow$ one matrix.

**Size drill:** write sizes side by side; inner match, outer = result.

## 10 · Linear systems

$Ax = b$: none / one / infinitely many. $m < n$ under-determined (wide), $m > n$ over-determined (tall) — shape alone decides nothing.

Ex: $x_1 - x_2 = 2,\ x_1 + x_2 = 4,\ 3x_1 - x_2 = 8$ → unique $(3, 1)$ although tall. $x = 1, x = 2$: none.

## 11 · Inverses

**Left** $XA = I \Rightarrow$ independent columns $\Rightarrow$ square or tall; $x = Xb$ is the **only** candidate (check it). Not unique.

**Right** $AX = I \Rightarrow$ independent rows $\Rightarrow$ square or wide; $x = Xb$ always solves; $X^\top$ is a left inverse of $A^\top$.

Both $\Rightarrow X = X(AY) = (XA)Y = Y = A^{-1}$ (square, unique). Square: left-inv $\Leftrightarrow$ indep cols $\Leftrightarrow$ right-inv $\Leftrightarrow$ indep rows $\Leftrightarrow \det \neq 0 \Leftrightarrow$ rank $n \Leftrightarrow \mathcal{N} = \{\mathbf{0}\}$.

$(A^{-1})^{-1} = A$, $(AB)^{-1} = B^{-1}A^{-1}$, $(A^\top)^{-1} = (A^{-1})^\top$, $(\alpha A)^{-1} = \alpha^{-1} A^{-1}$.

**2×2:** $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ — swap diagonal, negate off-diagonal, divide by det. **Check** $AA^{-1} = I$.

Diagonal: invertible iff all $d_i \neq 0$, inverse $\operatorname{diag}(1/d_i)$.

Vandermonde invertible iff distinct $t_i$. **Gram** $A^\top A$ **invertible $\Leftrightarrow$ independent columns** ($A^\top Ax = 0 \Rightarrow \|Ax\|^2 = 0 \Rightarrow x = 0$).

Orthonormal columns: $A^\top A = I$; square $\Rightarrow$ orthogonal, $Q^{-1} = Q^\top$, $\det Q = \pm 1$; preserves inner products, norms, distances, angles. Tall: $A^\top A = I$ but $AA^\top \neq I$.

Singular $\Rightarrow$ squashes some $x \neq 0$ to 0, info lost. Don't form $A^{-1}$ numerically (ill-conditioning).

## 12 · Determinant & transformations

$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$ = signed area of the column parallelogram. $\det = 0 \Rightarrow$ flattens $\Rightarrow$ singular; $< 0 \Rightarrow$ flip.

$\det(AB) = \det A \det B$, $\det A^\top = \det A$, $\det A^{-1} = 1/\det A$, $\det(cA) = c^n \det A$, triangular = product of diagonal, $= \prod \lambda_i$.

Rotation $\begin{pmatrix} \cos & -\sin \\ \sin & \cos \end{pmatrix}$ ($\det 1$); scaling $\operatorname{diag}(s_1, s_2)$; shear $\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$; reflection $\operatorname{diag}(1, -1)$; projection $\operatorname{diag}(1, 0)$ ($\det 0$).

## 13 · Pseudo-inverses

- Tall, indep cols: $A^\dagger = (A^\top A)^{-1} A^\top$ (left inverse) → **least squares**. Wide, indep rows: $A^\dagger = A^\top (AA^\top)^{-1}$ (right inverse) → **minimum-norm** solution.
- Moore–Penrose $A^+$ exists for every matrix; $= A^{-1}$ if invertible; $AA^+A = A$. $M \in \mathbb{R}^{5\times 2} \Rightarrow M^\dagger \in \mathbb{R}^{2\times 5}$.

## 14 · Rank

- Rank = #indep cols = #indep rows = #pivots $\leq \min(m, n)$. Full column rank $= n$; full row rank $= m$.
- $\operatorname{rank} A = \operatorname{rank} A^\top$; $\operatorname{rank}(AB) \leq \min(\operatorname{rank} A, \operatorname{rank} B)$ (can be strict: $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^2 = 0$); invertible $\Rightarrow$ rank $n$; $\operatorname{rank}(A^\top A) = \operatorname{rank} A$; $uv^\top$ rank 1.
- 3×3: rank 3 volume, 2 plane, 1 line, 0 origin.
- **Elimination:** swap rows, scale by non-zero, add multiple of a row. REF (pivots stair-step), RREF (pivots 1, alone in column). RREF shows redundant-column recipes.

## 15 · Column & null space, rank–nullity

- $\mathcal{C}(A) = \operatorname{span}(\text{cols}) = \{Ax\} \subseteq \mathbb{R}^m$, dim = rank; basis = **pivot columns of the original** $A$.
- $Ax = b$ solvable $\Leftrightarrow b \in \mathcal{C}(A) \Leftrightarrow \operatorname{rank} A = \operatorname{rank}[A|b]$ (zero rows of $[A|b]$ give the conditions).
- $\mathcal{N}(A) = \{x : Ax = \mathbf{0}\} \subseteq \mathbb{R}^n$; nullity = #free vars; special solutions: free var = 1, others 0.
- **Rank–nullity:** $\operatorname{rank} A + \dim \mathcal{N}(A) = n$ (**columns**). Wide $\Rightarrow$ non-trivial null space. $\mathcal{N}(A) = \{\mathbf{0}\} \Leftrightarrow$ indep cols $\Leftrightarrow$ rank $n$.
- Left null space $\mathcal{N}(A^\top) \subseteq \mathbb{R}^m$, dim $m - r$; row space dim $r$.
- **Complete solution** $x = x_p + x_n$ ($x_p$: free vars = 0). Precisely: $r = n \Rightarrow$ at most one; $r < n \Rightarrow 0$ or $\infty$; $r < m \Rightarrow$ some $b$ unreachable.
- Cases: $r = m = n$ one; $r = n < m$ 0 or 1; $r = m < n$ $\infty$ for every $b$; $r < m, n$ 0 or $\infty$.
- Duplicate feature (cm & inches) $\Rightarrow \mathcal{N}(X) \neq \{0\}$, $X^\top X$ singular, weights not unique.

## 16 · Supervised learning

- $D = \{(x_i, y_i)\} \subseteq \mathcal{X}^d \times \mathcal{Y}$; feature space, label space. Regression $\mathcal{Y} = \mathbb{R}$; binary $\{0, 1\}$; $M$-class $\{1..M\}$. Probability of X → regression; whether X → classification; no labels → neither.
- Unknown $P(X, Y)$; hypothesis $h \in \mathcal{H}$ (hypothesis class); learning $h^* = \arg\min_{h \in \mathcal{H}} \mathcal{L}(h)$.
- Losses ($\div n$, $\geq 0$): 0/1 $\frac{1}{n}\sum(1 - \delta_{h(x_i) - y_i})$ = misclassification rate (non-differentiable); MSE $\frac{1}{n}\sum e_i^2$; RMSE $\sqrt{\text{MSE}}$ (units of $y$); MAE $\frac{1}{n}\sum |e_i|$ (robust).
- Best constant: MSE → **mean**; MAE → **median**.
- Memoriser: 0 train error, useless $\Rightarrow$ overfitting. Split $D_{TR}$ (70–90%) / $D_{TE}$; i.i.d. or by time; **test once**. Train = parameters; validation = hyper-parameters; test = final assessment. $K$-fold CV loss $= \frac{1}{K}\sum \ell_j$.
- Classifier map: frequency table (Bayes, trees); covariance (LDA, logistic); similarity (**kNN**); others (NN, SVM).

## 17 · kNN

- Vote among the $k$ nearest; regression: mean or $\sum(y_i/d_i)/\sum(1/d_i)$. Lazy, non-parametric, no distribution assumption; hyper-params $k$, metric.
- Odd $k$ for binary (3 classes can still tie); ties → priors or $(k-1)$/1-NN; missing values → feature mean; categorical → **Hamming** (#mismatches).

Minkowski $p = 1, 2, \infty$. **Normalise** (min–max $\frac{x-\min}{\max-\min}$ or z-score), using training stats — scaling can flip the answer. $k = 1$: noise-sensitive, high variance, 0 train error; $k = n$: majority class. Larger $k \Rightarrow$ ↓variance ↑bias; non-linear boundary ⇒ small $k$; pick $k$ by CV.

Cost per query: distances $O(dn)$, select $O(n)$, vote $O(k) \Rightarrow O(dn)$ time & storage; training free.

**Curse of dimensionality:** window side 0.1 holds $0.1^p$ of data; to hold 10% need $L = 0.1^{1/p}$ ($p = 100$: 0.977).

## 18 · Classifier metrics (rows = predicted, cols = actual)

TP ($y{=}1, \hat{y}{=}1$), TN, FP (false alarm, $y{=}0, \hat{y}{=}1$), FN (miss). $P = TP + FN$, $N = FP + TN$ (column totals).

Acc $\frac{TP+TN}{P+N}$; Err $\frac{FP+FN}{P+N}$; **Recall/TPR/Sens** $\frac{TP}{P}$; **Spec/TNR** $\frac{TN}{N}$; **FPR** $\frac{FP}{N} = 1-\text{Spec}$; **Precision** $\frac{TP}{TP+FP}$; **NPV** $\frac{TN}{TN+FN}$.

Heart: TP100 FP10 FN5 TN50 → Acc .91, Sens .95, Spec .83, FPR .17, Prec .91, NPV .91, F1 .93, MCC .80.

Accuracy misleads under imbalance ("always legal" no-ball = 95%). Precision depends on prevalence.

Disease: avoid FN → **recall** (want FP ≫ FN). Fraud/spam: avoid FP → **specificity, precision**.

All-positive: Sens 1, Spec 0, Prec $P/(P+N)$. All-negative: Spec 1, Sens 0. Threshold trades them.

**ROC**: (FPR, TPR) over thresholds; ideal $(0, 1)$; diagonal = random; **AUC** = P(random + outscores random −); 1 perfect, .5 random; AUC $a < .5$ → flip → $1 - a$. ROC hides FP when TN is huge.

$F_\beta = \dfrac{1+\beta^2}{1/P + \beta^2/R}$; $F_1 = \dfrac{2PR}{P+R} = \dfrac{2TP}{2TP+FP+FN}$; $\beta > 1$ favours recall. min ≤ HM ≤ GM ≤ AM ≤ max.

**MCC** $= \dfrac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} \in [-1, 1]$, uses all 4 cells, class-symmetric; F1 changes if you swap which class is positive.

Reconstruct: $P, N$ from prevalence → TP, FN from recall → FP from precision ($TP/(TP+FP)$) or spec → TN.

## 19 · Linear regression

$y = f(x) + n$; model $\hat{y} = \hat{f}(x, \theta)$ (structure + parameters); error $e = y - \hat{y}$. Recipe: model → loss → optimisation → solve (analytic/numeric).

$\hat{y} = \theta_0 + \theta^\top x$ (hyperplane, $d+1$ params). Design matrix $X = [\mathbf{1}\ \ x_i^\top] \in \mathbb{R}^{n \times (d+1)}$, $\hat{y} = Xw$.

LSE, MSE, RMSE share one minimiser. $\mathcal{L}(w) = \frac{1}{2}\|y - Xw\|^2 = \frac{1}{2}(y^\top y - 2w^\top X^\top y + w^\top X^\top X w)$.

Gradients: $\nabla a^\top x = a$, $\nabla x^\top x = 2x$, $\nabla x^\top P x = 2Px$ ($P$ sym), $\nabla(\frac{1}{2}x^\top A x + b^\top x) = Ax + b$.

**Normal equations** $X^\top X w = X^\top y$, $w = (X^\top X)^{-1} X^\top y$. By projection: $X^\top(y - Xw) = \mathbf{0}$ (residual ⊥ $\mathcal{C}(X)$). Hessian $X^\top X$ PSD ⇒ global min.

Checks: residuals sum to 0; line passes through $(\bar{x}, \bar{y})$. 1 feature: $w_1 = S_{xy}/S_{xx}$, $w_0 = \bar{y} - w_1\bar{x}$. Bias only ⇒ $\theta_0 = \bar{y}$. Symmetric $x$ ⇒ $X^\top X$ diagonal, decoupled.

Fails if $n < d+1$ or redundant features ⇒ remove or regularise. Use `lstsq`, standardise (condition number), cost $n(d+1)^2 + (d+1)^3$.

Age data: $X^\top X = \begin{pmatrix} 4 & 145 \\ 145 & 5625 \end{pmatrix}$, $X^\top y = (14, 470) \Rightarrow w = (7.186, -0.1017)$.

## 20 · Polynomial regression & fitting

- $\hat{f} = \sum_{i=0}^{M} \theta_i x^i = g(x)^\top \theta$, $g = [1, x, \ldots, x^M]$: **linear in parameters**; $X$ Vandermonde; any fixed $g$ (sin, $e^x$) works; $w_0 + e^{w_1 x}$ does not.
- $M$ = hyper-parameter (like $k$). Bishop $\sin(2\pi x)$, $n = 10$: $M = 0, 1$ underfit; $M = 3$ good; $M = 9$ zero train error, wild, coefficients explode.
- $M \geq n - 1$ ⇒ exact interpolation. Remedies: (1) validation ( $E_{RMS}$ vs $M$, LOO for tiny data); (2) more data; (3) regularisation.
- Nested models: bigger model never worse on **training**; on **test** depends on bias vs variance.

## 21 · Ridge, lasso, elastic net

- $\mathcal{L}_{reg} = \frac{1}{2}\|y - X\theta\|^2 + \lambda\mathcal{R}(\theta)$; $\lambda = 0$ OLS, $\lambda \to \infty \to \mathbf{0}$ (constant $\bar{y}$).
- **Ridge** $\frac{\lambda}{2}\|\theta\|^2$: $\nabla = X^\top X\theta - X^\top y + \lambda\theta \Rightarrow \theta = (X^\top X + \lambda I)^{-1} X^\top y$. Always invertible ($v^\top(X^\top X + \lambda I)v \geq \lambda\|v\|^2 > 0$ ); unique; duplicates split equally.
- 1-D: $w = \dfrac{\sum x_n y_n}{\sum x_n^2 + \lambda}$; centred $w_1 = \dfrac{S_{xy}}{S_{xx} + \lambda}$ = shrink factor $\frac{S_{xx}}{S_{xx}+\lambda}$ × OLS. ↑λ: ↑bias ↓variance. Never exactly 0. Weight decay; Gaussian prior.
- Don't penalise intercept; **standardise first** (not unit-free). Geometry: disc. Use $\ln\lambda$; choose by validation (U-shaped curve).
- **Lasso** $\lambda\|\theta\|_1$: no closed form (coordinate descent), costlier; diamond corners ⇒ **sparse** (feature selection), biased. 1-D: $w = \dfrac{\text{sign}(b)\max(|b| - \lambda, 0)}{a}$ vs ridge $\dfrac{b}{a+\lambda}$, OLS $\dfrac{b}{a}$ ($a = \sum x^2$, $b = \sum xy$). Zero iff $|b| \leq \lambda$.
- **Elastic net** $\frac{1}{2}\|\cdot\|^2 + \lambda_1\|\theta\|_1 + \frac{\lambda_2}{2}\|\theta\|^2$; ridge & lasso special cases; 2 hyper-params.
- ↓variance: more data, simpler model, ↑λ. Sparse ⇒ $L^1$.

## 22 · Eigenvalues & eigenvectors

- $Aq = \lambda q$, $q \neq \mathbf{0}$, square $A$; only scaling (stretch / shrink / flip / squash). Defined up to scale.
- $\det(A - \lambda I) = 0$ (degree $n$); **2×2:** $\lambda^2 - \text{tr}(A)\lambda + \det A = 0$. $\sum\lambda_i = \text{tr}$, $\prod\lambda_i = \det$. Eigenspace $= \mathcal{N}(A - \lambda I)$; spectrum = set of $\lambda$.
- **Eigenvector recipe:** plug $\lambda$ into $A - \lambda I$, take **one row**, solve for one entry, set the other to 1, normalise, **check** $Aq = \lambda q$.
- **Symmetric shortcut:** eigenvectors for distinct $\lambda$ are ⊥ — for 2×2, if $q_1 = (a, b)$ then $q_2 = (-b, a)$.
- $AQ = Q\Lambda \Rightarrow A = Q\Lambda Q^{-1}$ (needs independent eigenvectors; distinct $\lambda$ suffice). Order of $\lambda$'s matches columns of $Q$.
- Meaning: $Q^{-1}$ change basis → $\Lambda$ scale → $Q$ back; $Ax = \sum\lambda_i c_i q_i$, $x = \sum c_i q_i$. Unit circle → ellipse, semi-axes $|\lambda_i|$ along $q_i$, area $|\det|\pi$.
- $\det A = \prod\lambda_i$; $A^{-1} = Q\Lambda^{-1}Q^{-1}$ (eigs $1/\lambda_i$); $A^k = Q\Lambda^k Q^{-1}$; $f(A)$ has eigs $f(\lambda_i)$ (e.g. $4A^3 + 6A^{-1} + 2I$); $A + cI$ shifts by $c$.
- $\lambda = 0 \Leftrightarrow$ dependent cols $\Leftrightarrow \det 0 \Leftrightarrow$ singular. Negative λ fine (flip). Repeated $A^k \to$ direction of largest $|\lambda|$.
- **Spectral theorem:** $A = A^\top \Rightarrow$ real λ, orthonormal eigenvectors, $A = Q\Lambda Q^\top$ ($Q^{-1} = Q^\top$), $A = \sum\lambda_i q_i q_i^\top$. Rotation $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$: $\lambda^2 + 1$, no real λ.

$\begin{pmatrix} a & b \\ b & a \end{pmatrix}$: eigs $a \pm b$, vectors $(1,1)$, $(1,-1)$. E.g. $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$: 3, 1; $C^2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$; $S^k = \frac{1}{2}\begin{pmatrix} 3^k+1 & 3^k-1 \\ 3^k-1 & 3^k+1 \end{pmatrix}$.

## 23 · Preview (insurance): PSD, SVD

PSD $\Leftrightarrow x^\top Ax \geq 0 \Leftrightarrow$ all $\lambda \geq 0 \Leftrightarrow A = B^\top B$; $\det > 0$ alone insufficient. $X^\top X$ PSD; $+\lambda I \Rightarrow$ PD.

SVD $A = U\Sigma V^\top$; $V$ eigvecs of $A^\top A$, $U$ of $AA^\top$, $\sigma_i = \sqrt{\lambda_i(A^\top A)}$, $Av_i = \sigma_i u_i$; $\det(A^\top A) = \prod \sigma_i^2$.

Rank-$k$ $A_k = \sum_{i \leq k} \sigma_i u_i v_i^\top$, $\|A - A_k\|_F^2 = \sum_{i>k} \sigma_i^2$; storage $mk + k + kn$ vs $mn$.

## 24 · Worked procedures (exam-style)

**2×2 inverse.** $M = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$, $\det = 6 - 5 = 1$, $M^{-1} = \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$; $Mx = (4,7) \Rightarrow x = M^{-1}(4,7) = (1,1)$. $\begin{pmatrix} 4 & 7 \\ 2 & 6 \end{pmatrix}^{-1} = \frac{1}{10}\begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix}$.

**Symmetric eigenvectors.** $A = \begin{pmatrix} 3 & -2 \\ -2 & 6 \end{pmatrix}$: $\lambda^2 - 9\lambda + 14 = 0 \Rightarrow 7, 2$. $\lambda = 7$: row 1 of $A - 7I$: $-4a - 2b = 0 \Rightarrow q_1 = (1,-2)/\sqrt{5}$; $\perp \Rightarrow q_2 = (2,1)/\sqrt{5}$ (check $Aq_2 = 2q_2$ ✓). $A = Q\Lambda Q^\top$, $A^3 = Q\,\mathrm{diag}(343, 8)Q^\top = \begin{pmatrix} 75 & -134 \\ -134 & 276 \end{pmatrix}$, $A^{-1} = \frac{1}{14}\begin{pmatrix} 6 & 2 \\ 2 & 3 \end{pmatrix}$ (eigs $\frac{1}{7}, \frac{1}{2}$).

$\begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$: $\lambda^2 - 7\lambda + 6 \Rightarrow 6, 1$; $q_1 = (2,1)/\sqrt{5}$, $q_2 = (-1,2)/\sqrt{5}$. $\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ (not sym.): 5, 2; $(1,1)$, $(1,-2)$ — not $\perp$.

**Gram–Schmidt in $\mathbb{R}^3$.** $a_1 = (2,1,2)$, $\|a_1\| = 3$, $q_1 = \frac{1}{3}(2,1,2)$; $a_2 = (1,1,0)$, $q_1^\top a_2 = 1$, $\tilde{q}_2 = a_2 - q_1 = \frac{1}{3}(1,2,-2)$, $\|\cdot\| = 1 \Rightarrow q_2 = \frac{1}{3}(1,2,-2)$. Project $b = (0,0,3)$: $(q_1^\top b)q_1 + (q_2^\top b)q_2 = 2q_1 - 2q_2 = \frac{2}{3}(1,-1,4)$; residual $\frac{1}{3}(-2,2,1) \perp q_1, q_2$.

**Complete solution.** Reduce $[A|b]$ to RREF; pivots → rank $r$; free vars $n - r$. $x_p$: free = 0, read pivots. Special solutions: each free var = 1 in turn, back-substitute with $b = 0$. $x = x_p + \sum c_i s_i$. E.g. $x = (2,0,3,0) + c_1(-2,1,0,0) + c_2(-1,0,-2,1)$; check $As_i = \mathbf{0}$.

**Least squares (1 feature).** Data $(0,1), (1,2), (2,2), (3,4)$: $\bar{x} = 1.5$, $\bar{y} = 2.25$, $S_{xx} = 5$, $S_{xy} = 4.5 \Rightarrow w_1 = 0.9$, $w_0 = 2.25 - 1.35 = 0.9$. Matrix form: $X^\top X = \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix}$, $X^\top y = (9, 18)$, $w = (X^\top X)^{-1}X^\top y = \frac{1}{20}\begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix}(9,18) = (0.9, 0.9)$.

Ridge (centred, $\lambda = 5$): $w_1 = 4.5/10 = 0.45$, $w_0 = \bar{y} - w_1\bar{x} = 1.575$.

**Lasso / soft-threshold (1-D).** $a = \sum x^2$, $b = \sum xy$: OLS $b/a$, ridge $b/(a+\lambda)$, lasso $\mathrm{sign}(b)(|b| - \lambda)_+/a$. $a = 8, b = -12, \lambda = 4$: OLS $-1.5$, ridge $-1$, lasso $-1$; $\lambda \geq 12 \Rightarrow$ lasso 0.

**kNN by hand.** Normalise if scales differ → distances to every training point (square-root optional for ranking) → sort → vote for $k = 1, 3, 5$ (answers can change with $k$) → report ties.

**Metrics by hand.** Build the table (rows predicted, cols actual); write $P$, $N$; then Acc, Recall, Spec, Prec, F1. TP30 FN10 FP20 TN140: Acc .85, Rec .75, Spec .875, Prec .6, F1 .667, FPR .125.

**Correlation by hand.** Centre both vectors, $\rho = \tilde{a}^\top\tilde{b}/(\|\tilde{a}\|\|\tilde{b}\|)$. $a = (1,2,3)$, $b = (2,4,7)$: $\tilde{a} = (-1,0,1)$, $\tilde{b} = (-2.33, -0.33, 2.67) \Rightarrow \rho = 5/(\sqrt{2}\sqrt{12.67}) = 0.993$.

**Projection by hand.** $x = (3,4)$ on $u = (1,1)$: $\frac{7}{2}(1,1)$, residual $(-\frac{1}{2}, \frac{1}{2}) \perp u$; distance from $x$ to the line $= \frac{1}{\sqrt{2}}$.

## 25 · Exam drill & traps

- MCQ: tick only what you can justify (−1 each wrong). Problems: write formula → substitute → number → **check**.
- Checks: $q_i^\top q_j = 0$; residuals sum 0, $X^\top e = 0$; $\sum \lambda = \mathrm{tr}$, $\prod \lambda = \det$; $AA^{-1} = I$; confusion rows/cols sum to totals; rank + nullity = $n$.
- Traps: $\|x\|_2 \neq \#$ (it's √); rank–nullity counts **columns**; ridge never 0; lasso sparse; recall = TP/P not TP/(TP+FP); accuracy under imbalance; $(AB)^{-1} = B^{-1}A^{-1}$; $\det(cA) = c^n \det A$; wide ⇒ no left inverse, tall ⇒ no right inverse; GS order matters; MSE→mean, MAE→median.

## 26 · Quick true/false facts (MCQ bank)

- **T**: every subspace contains $\mathbf{0}$; span is a subspace; $\mathcal{N}(A)$, $\mathcal{C}(A)$ are subspaces; solution set of $Ax = b$, $b \neq 0$, is **not** a subspace (no $\mathbf{0}$).
- **T**: $n$ orthonormal vectors in $\mathbb{R}^n$ form a basis; orthogonal non-zero ⇒ independent (converse false).
- **T**: $A^\top A$ and $AA^\top$ have the same non-zero eigenvalues, both PSD, rank $= \mathrm{rank}\, A$.
- **T**: wide $A$ $(m < n) \Rightarrow \mathcal{N}(A) \neq \{0\} \Rightarrow Ax = b$ never unique. Tall $A \Rightarrow$ some $b$ have no solution.
- **F**: "tall ⇒ no solution" (can be consistent); "square ⇒ invertible"; "$\det A > 0 \Rightarrow$ PSD"; "independent ⇒ orthogonal"; "basis must be orthonormal".
- **F**: $\|u + v\| = \|u\| + \|v\|$ in general (only when same direction); $\|x\|_0$ is a norm; $\|x\|_\infty \geq \|x\|_2$.
- **T**: $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ even when $AB \neq BA$; $\det(AB) = \det(BA)$ for square.
- **T**: eigenvectors for distinct eigenvalues are independent; symmetric ⇒ orthogonal. Same eigenvalue ⇒ any combination is still an eigenvector.
- **T**: $A$ invertible ⇔ 0 is not an eigenvalue; $A^k$, $A^{-1}$, $A + cI$ share $A$'s eigenvectors.
- **T**: ridge ⇒ unique solution even with duplicate features; lasso ⇒ may zero features; both shrink; $\lambda$ chosen on validation, not test.
- **T**: increasing polynomial degree never increases training error; test error is U-shaped. $k\uparrow$ in kNN ⇒ smoother boundary.
- **T**: kNN has no training phase but $O(dn)$ prediction; needs feature scaling; suffers in high dimension.
- **T**: AUC is threshold-free; F1 ignores TN; MCC uses all four; precision undefined if no positives predicted.
- **T**: correlation is invariant to positive scaling & shifting of either vector; cosine similarity is invariant to scaling only.
- **T**: $\mathrm{std}(x + c\mathbf{1}) = \mathrm{std}(x)$, $\mathrm{std}(\alpha x) = |\alpha|\,\mathrm{std}(x)$; avg shifts by $c$.
- **T**: rotation/orthogonal $Q$: $\|Qx\| = \|x\|$, $\det = \pm 1$, columns & rows orthonormal; permutation is orthogonal.
- **T**: outer product $uv^\top$ $(u, v \neq 0)$ has rank 1, column space $= \mathrm{span}(u)$, null space $= v^\perp$ (dim $n - 1$).